

Lecture 11 Worksheet

Math 124, Fall 2026 at the University of Michigan

Some lectures will have an accompanying worksheet. We will do our best to include these problems directly in the notes as activities or worked examples.

Overall, how did you feel about Midterm 1?

Circle one.



Activity 1

Let

$$A = \begin{bmatrix} 4 & -6 & 9 \\ -3 & 7 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & 3 & -4 \\ 6 & -5 & 2 \end{bmatrix}.$$

a) Compute $A - 3B$.

b) If C has shape 3×2 , is $A + C$ defined? Explain.

c) What are the shape and entries of $0A$?

Activity 2

Let

$$A = \begin{bmatrix} 4 & -3 & 5 \\ -2 & 6 & 7 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

a) Compute $A\vec{v}$.

b) List different ways to interpret $A\vec{v}$. Explain how each interpretation describes this calculation.

c) Let

$$B = \begin{bmatrix} 3 & -4 & 8 \\ -5 & 2 & -3 \\ 6 & 7 & 4 \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}.$$

Compute $B\vec{u}$.

Activity 3

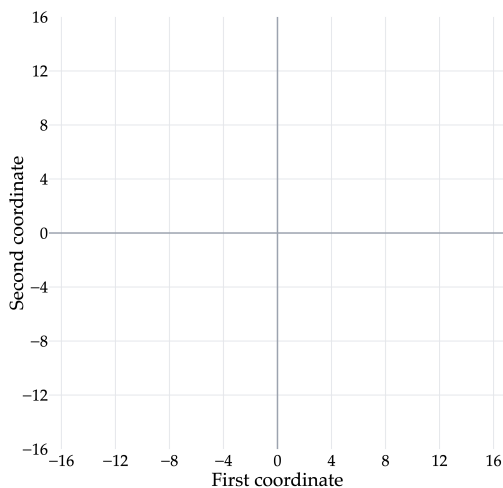
In your group of four, each person should pick a different example from Examples 1–5. Work on your chosen example individually first, then work on the remaining example together and discuss all five examples. Explain your calculations and what each matrix does to a vector.

Example 1

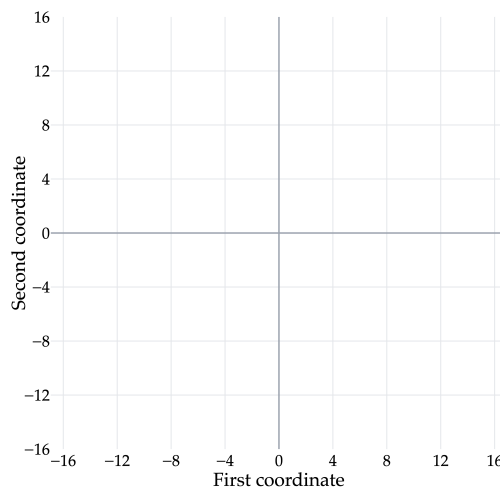
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}.$$

- a) Compute $A\vec{u}$ and $A\vec{v}$.

- b) Draw and label $\vec{u}, \vec{v}$ on the Before graph and $A\vec{u}, A\vec{v}$ on the After graph. Use arrows from the origin.



Before: $\vec{u}, \vec{v}$



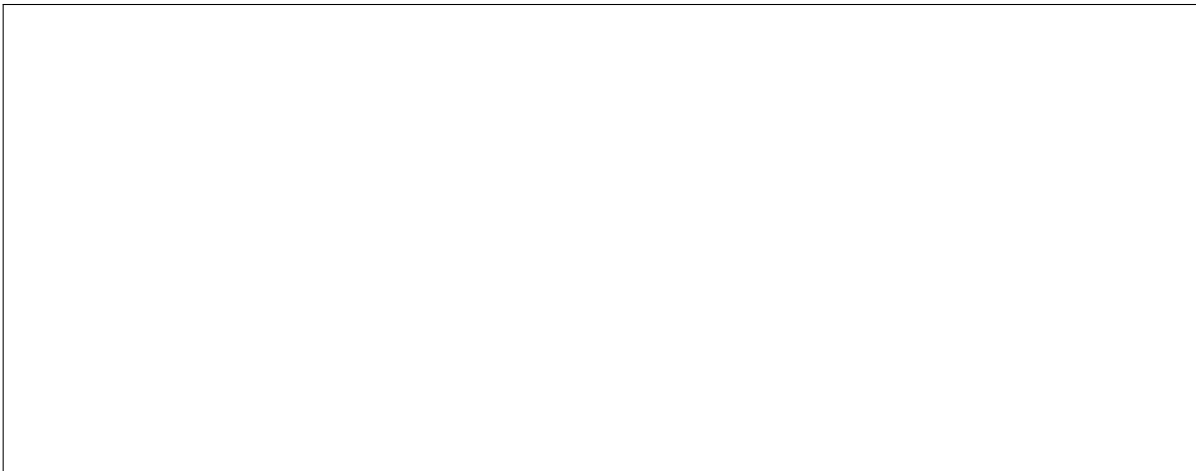
After: $A\vec{u}, A\vec{v}$

- c) Compute $A \begin{bmatrix} a \\ b \end{bmatrix}$, where a and b are scalars. Describe the operation, then record the name and definition of this type of matrix during the class discussion.

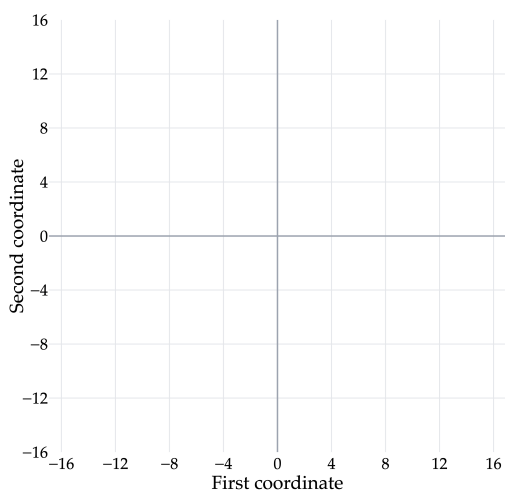
Example 2

$$A = \begin{bmatrix} -3/2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}.$$

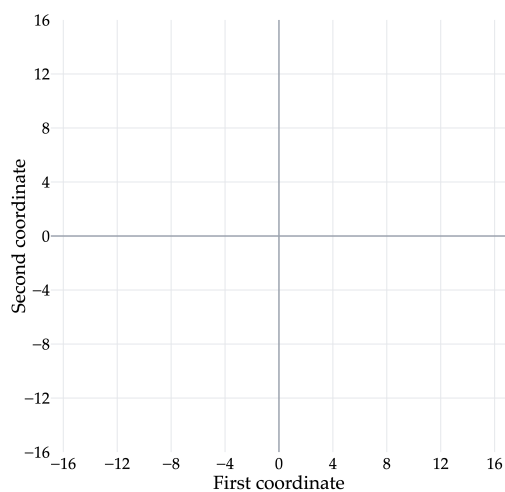
- a) Compute $A\vec{u}$ and $A\vec{v}$.



- b) Draw and label $\vec{u}, \vec{v}$ on the Before graph and $A\vec{u}, A\vec{v}$ on the After graph. Use arrows from the origin.



Before: $\vec{u}, \vec{v}$



After: $A\vec{u}, A\vec{v}$

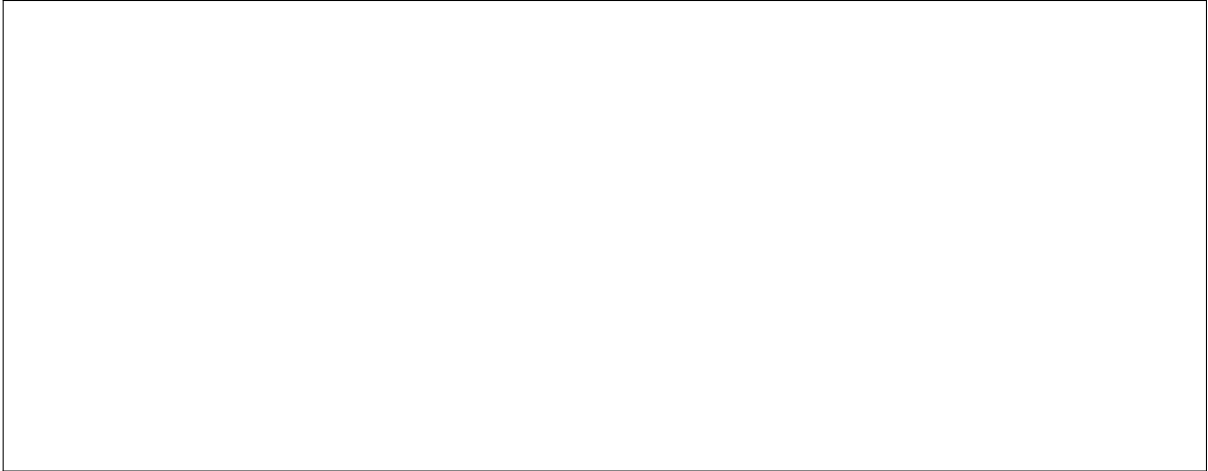
- c) Compute $A \begin{bmatrix} a \\ b \end{bmatrix}$, where a and b are scalars. Describe the operation, then record the name and definition of this type of matrix during the class discussion.



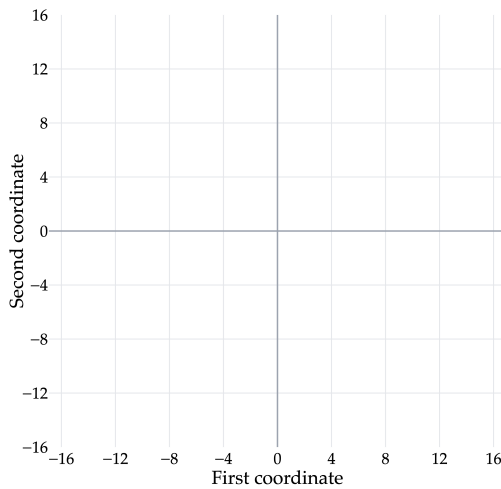
Example 3

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}.$$

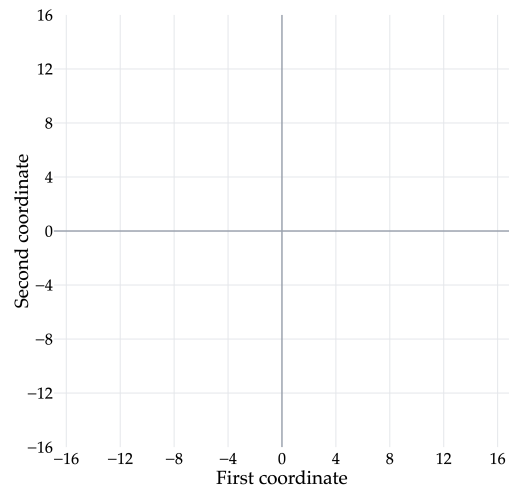
- a) Compute $A\vec{u}$ and $A\vec{v}$.



- b) Draw and label $\vec{u}, \vec{v}$ on the Before graph and $A\vec{u}, A\vec{v}$ on the After graph. Use arrows from the origin.



Before: $\vec{u}, \vec{v}$



After: $A\vec{u}, A\vec{v}$

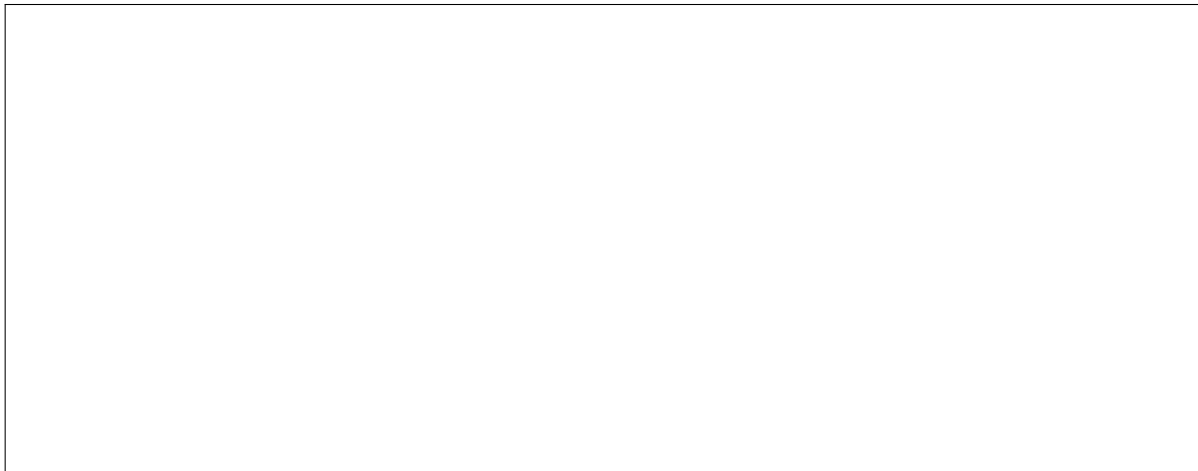
- c) Compute $A \begin{bmatrix} a \\ b \end{bmatrix}$, where a and b are scalars. Describe the operation, then record the name and definition of this type of matrix during the class discussion.



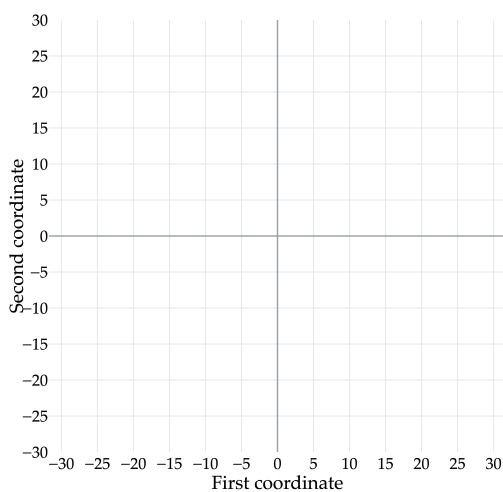
Example 4

$$A = \begin{bmatrix} 3 & 6 \\ -2 & -4 \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}.$$

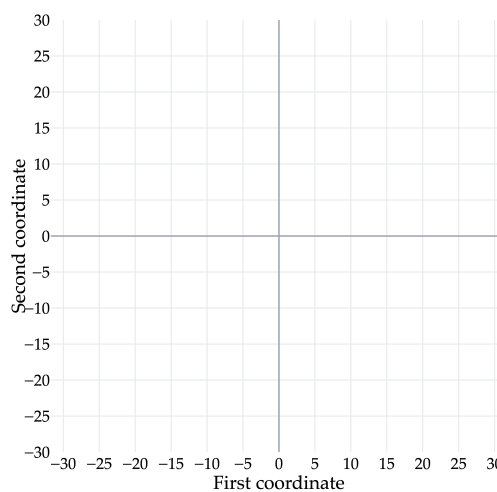
- a) Compute $A\vec{u}$ and $A\vec{v}$.



- b) Draw and label $\vec{u}, \vec{v}$ on the Before graph and $A\vec{u}, A\vec{v}$ on the After graph. Use arrows from the origin.



Before: $\vec{u}, \vec{v}$



After: $A\vec{u}, A\vec{v}$

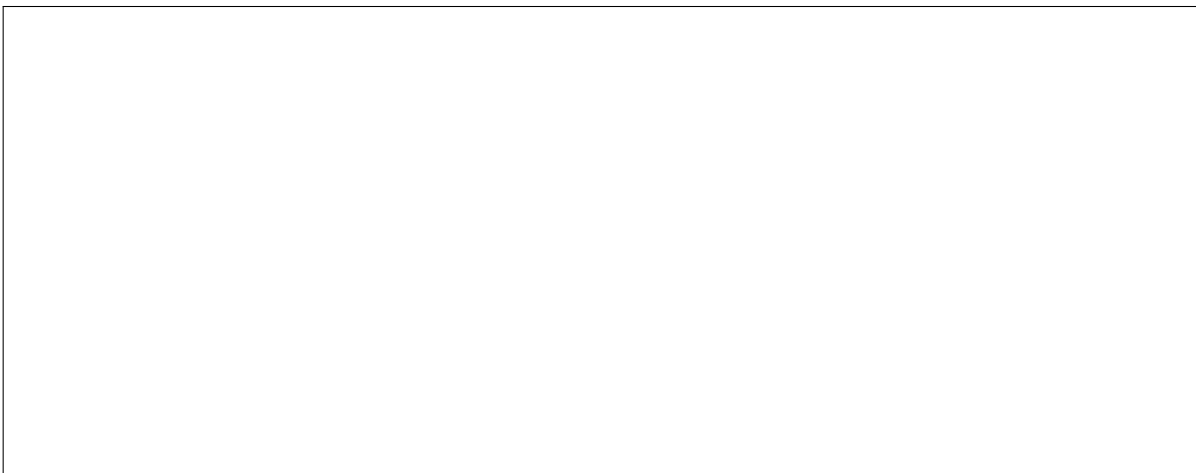
- c) Compute $A \begin{bmatrix} a \\ b \end{bmatrix}$, where a and b are scalars. Describe the operation. Explain why every output lies on the same line.



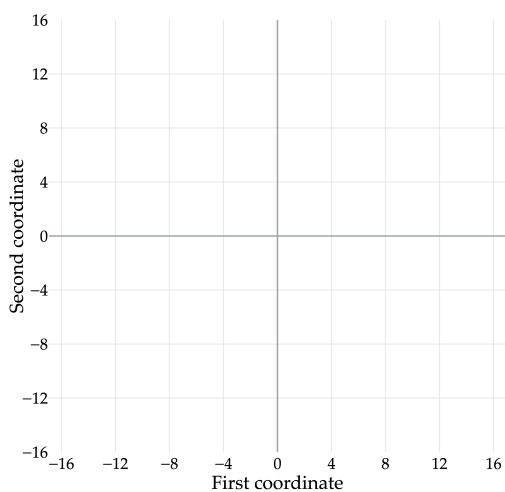
Example 5

$$A = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}.$$

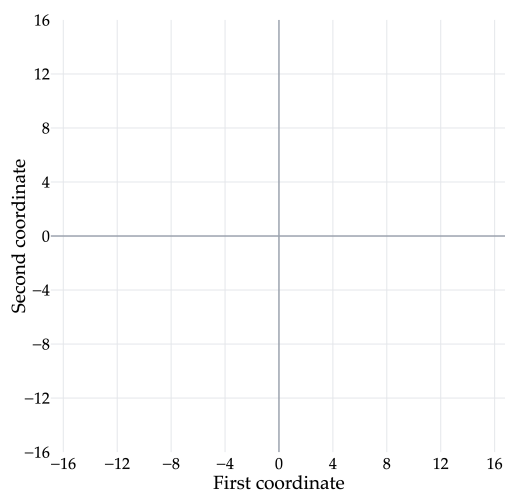
- a) Compute $A\vec{u}$ and $A\vec{v}$.



- b) Draw and label $\vec{u}, \vec{v}$ on the Before graph and $A\vec{u}, A\vec{v}$ on the After graph. Use arrows from the origin.



Before: $\vec{u}, \vec{v}$



After: $A\vec{u}, A\vec{v}$

- c) Compute $A \begin{bmatrix} a \\ b \end{bmatrix}$, where a and b are scalars. Describe the operation. Compare the lengths of the inputs and outputs. What changes, and what stays the same?

